Tallinn University

School of Digital Technologies

DTLGM.DT Digital Learning Games

Designing an XR Drone Training Simulator

Master's Thesis

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Abstract

Automation is progressively displacing skilled technical labor, however, current robotic systems remain limited in replicating complex tasks, leaving a persistent need for human operators. At the same time, such skills are difficult to acquire, requiring prolonged training and the development of muscle memory. This creates a gap between the continued demand for skilled operators and the accessibility of training pathways.

This paper presents a design-based research study investigating Extended Reality (XR) as a medium for lowering the learning barrier and increasing engagement in complex skill acquisition, using First Person View (FPV) quadcopter piloting as a representative case domain.

This study is structured around three primary objectives: (1) the simulation of FPV quadcopter flight physics. (2) the design of a human-AI co-training framework (3) the interaction instruments for a high-fidelity XR prototyping. In particular PID and RL (Reinforcement Learning) are used in the co-training AI, and Geographic Information System (GIS) data integrated to support the environmental representation.

The contribution of this work is a structured design framework that links physical simulation, AI-assisted co-training, and XR-based interaction design, aimed at reducing the learning curve and improving accessibility to complex skill training.

Keywords: Extended Reality, Digital Twin, FPV Drone, Human-AI Co-training, PID Control, Reinforcement Learning

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Introduction

Robots are increasingly capable of replacing routine manual and cognitive tasks. However, there remains a strong demand for human cognitive and motor skills that machines cannot yet replicate. At the same time, this technological shift is occurring alongside a growing labor shortage, partly because many workers are unwilling to take jobs that are physically demanding and are expected to be replaced by automation in the future.

As a result, industries such as agriculture, forestry, and heavy-duty mobile machinery are already facing significant shortages of skilled workers (Machado et al., 2021). Moreover, career expectations are increasingly influenced by perceptions of automation. Research shows that when individuals believe a job is likely to be replaced by artificial intelligence, their willingness to pursue that career decreases significantly (Wang et al., 2026).

Despite the growing level of automation, current robotic systems still struggle with complex and dynamic environments. Therefore, skilled human operators remain essential. However, acquiring these skills is difficult and time-consuming, as it requires extensive repeated practice to develop muscle memory.

In this context, learning theory provides a useful framework. Near transfer refers to applying skills in similar contexts, while far transfer refers to applying them in different contexts (Barnett & Ceci, 2002). For near transfer, repeated practice plays a critical role in building procedural knowledge and muscle memory.

Given these challenges, Extended Reality (XR) has emerged as a promising training approach. Previous studies have explored XR in domains such as medical procedures and aviation (Dutescu et al., 2025). Meanwhile, Unmanned Aerial Vehicles (UAVs) are widely used in construction, agriculture, logistics, and disaster management (Otto et al., 2018). Among these applications, First Person View (FPV) quadcopter piloting represents a particularly demanding skill that relies heavily on precise motor control and spatial awareness. Accordingly, this study uses FPV quadcopter piloting as a case to investigate XR-based training for muscle memory development.

However, existing drone simulators are primarily designed for entertainment rather than structured learning. Consequently, they often present steep learning curves and provide limited support for systematic skill acquisition. In addition, the lack of standardized controller configurations create further barriers (Ebeid et al., 2018), leading to user frustration even before effective training begins.

Taken together, low career expectations, high preparation effort, and early-stage frustration significantly reduce learner motivation. As a result, many beginners discontinue training before acquiring sufficient muscle memory.

To address these issues, the goal of this study is to design an XR-based training system that lowers the learning barrier for FPV quadcopter piloting. In particular, the proposed system aims to support muscle memory development through AI-assisted co-training and improved interaction design.

This study addresses the following research questions:

(1) How can FPV quadcopter flight physics be simulated?

(2) How can a human-AI collaboration agent be designed to assist users?

(3) What interaction methods enable high-fidelity XR prototyping?

This study uses a design-based research approach. It applies agile development practices and iterative prototyping to build an XR FPV quadcopter simulator, including physics simulation, AI co-training (PID control and reinforcement learning), and interaction design.

Theoretical Background

Introduction to Quadcopter Systems

A quadcopter, also frequently referred to as a quadrotor, is a type of multirotor helicopter or rotary-wing Unmanned Aerial Vehicle (UAV) that utilizes four rotors to achieve flight (Luukkonen, 2011). These rotors are typically directed upwards and arranged in a square formation, with each rotor positioned at an equal distance from the center of mass. In this configuration, the four actuators are generally equally spaced by 90° from one another (Idrissi et al., 2022).

The origins of quadrotor technology date back to 1907, when the Breguet brothers constructed the Gyroplane No. 1. While this marked one of the first manned helicopter flights, it was not a free flight; the machine lacked inherent stability and a proper control mechanism, requiring it to be tethered to the ground (Ghazbi et al., 2016).

The transition to modern UAVs occurred in the late 1990s. A pivotal moment was the 1999 release of the Draganflyer I, the first camera-equipped quadrotor to achieve mass production. This shifted quadrotor technology from a laboratory curiosity to a viable commercial product (Karahan, 2025).

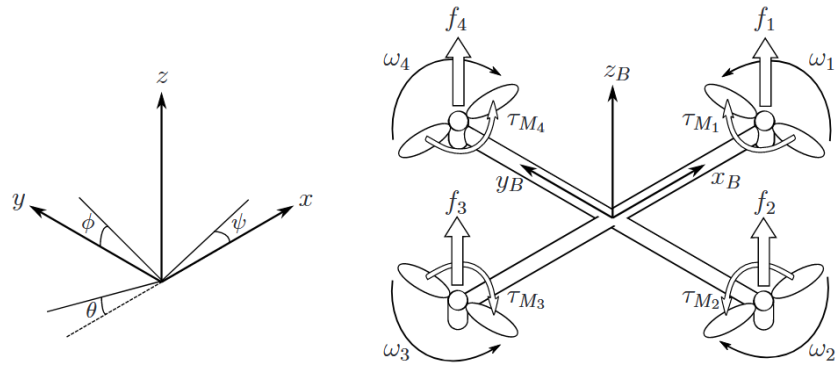
Since 2010, advancements in micro-electronics and control theory have expanded the use of quadrotors across various sectors:

- Public Safety: Search and rescue, firefighting, and border surveillance.
- Agriculture: Precision seeding, chemical spraying, and crop monitoring.
- Commercial: Last-mile delivery and high-end cinematography.

Flight Dynamics and Principles of Operation

A quadcopter is an under-actuated system, meaning it must manage six degrees of freedom (6DOF) as shown in Figure 1, three translational and three rotational—using only four control inputs (the angular velocities of the rotors).

Figure 1
The inertial and body frames of a quadcopter



The fundamental principle of quadcopter stability relies on torque cancellation. To maintain a stable orientation, two motors rotate clockwise (CW) while the adjacent pair rotates counter-clockwise (CCW). This arrangement ensures that the net moment about the center of mass is zero; without this pair-wise opposition, the drone would rotate uncontrollably around its vertical axis. Quadcopters typically use either a plus (+) or cross (x) configuration, with the cross configuration being more common due to its superior stability (Idrissi et al., 2022).

The absolute linear position of the quadcopter is defined in the inertial frame axes ($[x, y, z]^T$). The attitude, i.e.the angular position, is defined in the inertial frame with three Euler angles ($[\phi, \theta, \psi]^T$), which represent pitch angle, roll angle and yaw angle as shown in Figure 1.

Every change in position or attitude is achieved by varying the individual angular velocities (ω_i) of the four rotors:

Thrust (T): Simultaneously increasing all rotor speeds causes the vehicle to ascend. The force generated by rotor i is $f_i = k\omega_i^2$, where k is the lift constant. Total thrust is defined as show in Equation (1)

$$T = k \sum_{i=1}^{4} \omega_i^2 \tag{1}$$

Yaw (τ_ψ): Rotating the drone around the vertical axis is achieved by creating a torque imbalance between the CW and CCW pairs shown in Equation (2), where b is the drag constant.

$$\tau_\psi = b(\omega_1^2 - \omega_2^2 + \omega_3^2 - \omega_4^2) \tag{2}$$

Roll (τ_ϕ) and Pitch (τ_θ): These are achieved by creating a speed differential between opposite rotors. For a distance l from the center of mass, shown in Equation (3)

$$\begin{aligned} \tau_\phi &= lk(-\omega_2^2 + \omega_4^2) \\ \tau_\theta &= lk(-\omega_1^2 + \omega_3^2) \end{aligned} \tag{3}$$

The quadcopter is modeled as a rigid body using Newton-Euler equations. The acceleration in the inertial frame $[x, y, z]^T$ is governed by gravity (g) and the orientation of the thrust vector shown in Equation (4).

$$\begin{bmatrix} \ddot{x} \\ \ddot{y} \\ \ddot{z} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -g \end{bmatrix} + \frac{T}{m} \begin{bmatrix} \cos\psi \sin\theta \cos\phi + \sin\psi \sin\phi \\ \sin\psi \sin\theta \cos\phi - \cos\psi \sin\phi \\ \cos\theta \cos\phi \end{bmatrix} \tag{4}$$

In the body-fixed frame, the angular acceleration is influenced by the external torque (τ), inertia matrix (I), and gyroscopic effects as shown in Equation (5)

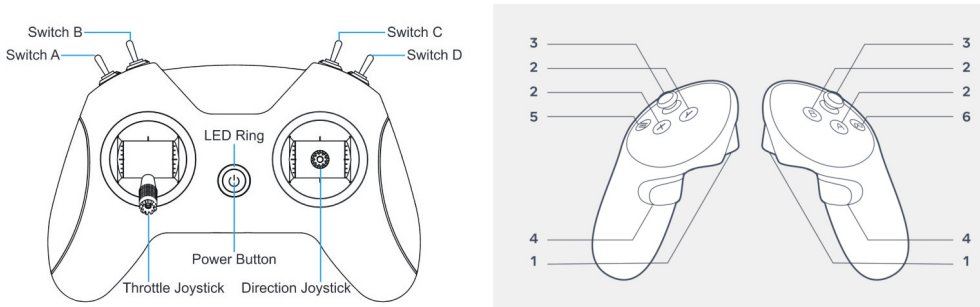
$$\begin{bmatrix} \dot{p} \\ \dot{q} \\ \dot{r} \end{bmatrix} = \begin{bmatrix} (I_{yy} - I_{zz})qr/I_{xx} \\ (I_{zz} - I_{xx})pr/I_{yy} \\ (I_{xx} - I_{yy})pq/I_{zz} \end{bmatrix} + \begin{bmatrix} \tau_\phi/I_{xx} \\ \tau_\theta/I_{yy} \\ \tau_\psi/I_{zz} \end{bmatrix} \tag{5}$$

Pilot Interface and Control Systems

The First-Person View (FPV) experience is defined by the synchronization between the pilot's inputs and the drone's visual feedback. An FPV drone can be controlled with a traditional remote controller or with a Motion Controller shown in Figure 2. A Motion Controller allows they user to maneuver the drone using hand gestures which is easy and convenient. The FPV Goggles is a also used for the FPV Drone. It's the defining factor for the FPV Drone. It connects to the drone's camera so that the camera user can have full vision of the quadcopter's camera

(Usama et al., 2021).

Figure 2
Transmitter (Left) and Gamepad (Right)



Adapted from (BetaFPV, 2022)

The industry standardizes these controls into two primary layouts: Mode 1 (Japanese Hand): Predominantly used in Japan and parts of Europe. The throttle is located on the right gimbal; Mode 2 (American Hand): The global standard for FPV.The throttle is located on the left gimbal, which most pilots find more intuitive for separating altitude from directional pitch. Difference between Mode 1 and Mode 2 is shown in Table 1

From a signal processing perspective, the flight controller differentiates between the unipolar throttle and the bipolar command sticks: Throttle (T): Mapped to a range of $[0,1]$. Roll, Pitch, and Yaw (ϕ, θ, ψ): Mapped to a range of $[-1,1]$. In Figure 3 shows the interaction of the transmitter in Mode 2 and how the quadcopter move.

FPV drones generally operate across two flight logic modes. Self-Leveling (Angle/Normal Mode): The onboard flight controller uses accelerometers to automatically return the drone to a level horizon when the sticks are centered. This is ideal for beginners but limits maneuverability; Acrobat (Manual Mode): The controller only uses gyroscopes to maintain the current angular rate. If the pilot moves the stick and returns it to center, the drone maintains its current tilted orientation. This allows for flips and rolls but requires constant manual correction.

The FPV goggles provide the pilot with a low-latency first person view video feed pass from the camera set on the quadcopter's front. Beyond the video feed, an On-Screen Display

Table 1
Comparison of Transmitter Control Modes

Mode	Left Stick (V)	Left Stick (H)	Right Stick (V)	Right Stick (H)
Mode 1	Pitch	Yaw	Throttle	Roll
Mode 2	Throttle	Yaw	Pitch	Roll

(OSD) provides critical telemetry data, including battery voltage, signal strength, and flight time, as shown in Figure 4.

Proportional Integral and Derivative (PID Control)

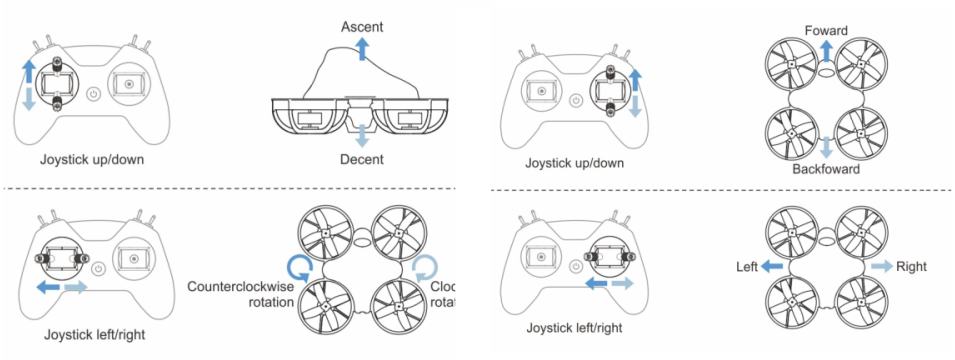
In 1922, Nicolas Minorsky presented a very clear analysis of the control actions necessary to provide effective control of a system whose exact dynamics were un- known. He analyzed the actions taken by a good helmsman steering a ship and translated these actions into the appropriate mathematical formulations (Bennett, 1993). It is now know as the Proportional-Integral-Derivative (PID) controller. PID is a type of controller that is most commonly used and applied in mechanical systems (Idrissi et al., 2022). Also used in many modern quadcopters, PID is employed here to stabilize drone movement and to function as a trainer AI.

The controller consists of three main components: Proportional (K_p): Acts like a virtual spring that pulls the system toward the desired position; the control action increases as the error gets larger; Integral (K_i): Accumulates the error over time to eliminate steady-state error, ensuring the drone reaches the exact target even in the presence of disturbances like gravity; Derivative (K_d): Acts as a virtual damper to reduce oscillations and prevent the system from overshooting its target.

The idealized control law is expressed as Equation (6).

$$u(t) = K_p e(t) + K_i \int e(t)dt + K_d \frac{de(t)}{dt} \tag{6}$$

Figure 3
Mode 2 control system



Adapted from (BetaFPV, 2022)

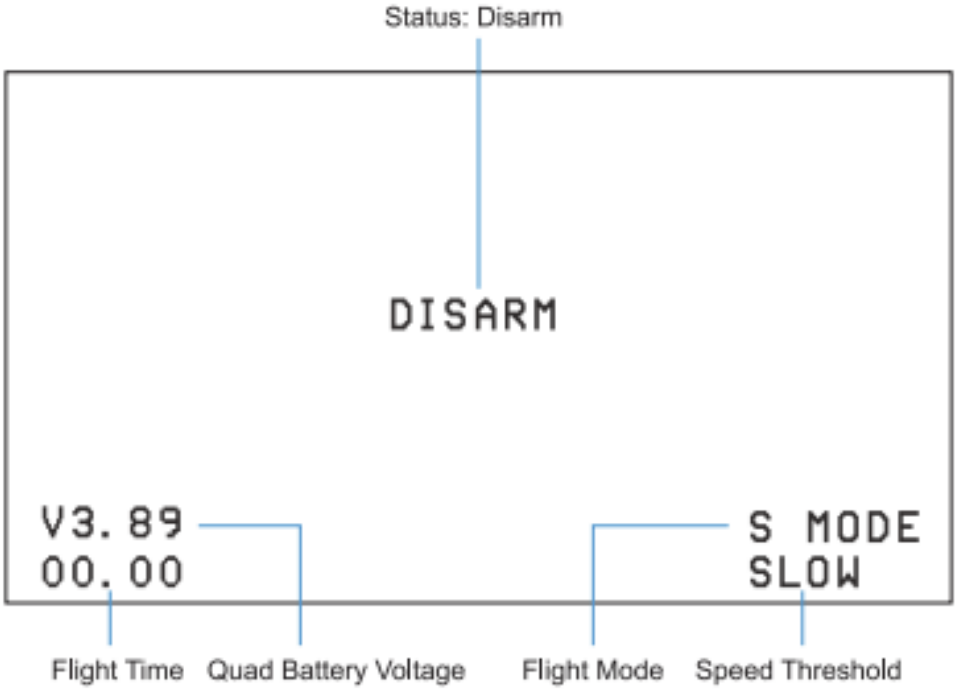
Learning Theories in Simulation-Based XR

Experiential Learning

Experiential learning theory posits that learning is a process through which knowledge is created via the transformation of experience (Kolb, 1984). This perspective emphasizes that learning emerges from the integration of experiencing, reflecting, thinking, and acting within a continuous cycle.

Kolb's model conceptualizes learning as a four-stage cyclical process. First, concrete experience involves the learner actively engaging in a specific activity, where technologies such as extended reality may enhance immersion. Second, reflective observation requires the learner to consciously reflect on the experience, often supported by guidance from a co-trainer or feedback mechanisms, in order to interpret its meaning. Third, abstract conceptualization refers to the development or integration of theoretical constructs or hypotheses derived from reflection. Finally, active experimentation involves applying these conceptualizations to new situations, enabling the learner to test their understanding and generate new experiences, thereby continuing the cycle.

Figure 4
The FPV goggles interface



Adapted from (BetaFPV, 2022)

Training Simulator

A training simulator can be defined as a system designed to replicate real-world processes or environments in order to facilitate skill acquisition and performance improvement (Narayanasamy et al., 2006). In educational contexts, it is often associated with simulation-based learning, which emphasizes immersive, experiential environments that enable learners to actively engage in realistic problem-solving scenarios (Dai & Ke, 2022). Furthermore, when game elements are incorporated, training simulators may also be conceptualized as serious games, which are designed with explicit educational or training purposes rather than pure entertainment (Caballero-Hernández et al., 2017).

Learning in Extended Reality (XR)

(Emma, 2026) (Lowell, 2024) (Curran et al., 2022) (Curran & Hollett, 2024) (Chen et al., 2022) (Zwoliński et al., 2022)

Human-AI Collaboration

(Yin et al., 2023) (Gao et al., 2025)

Reinforcement Learning (RL)

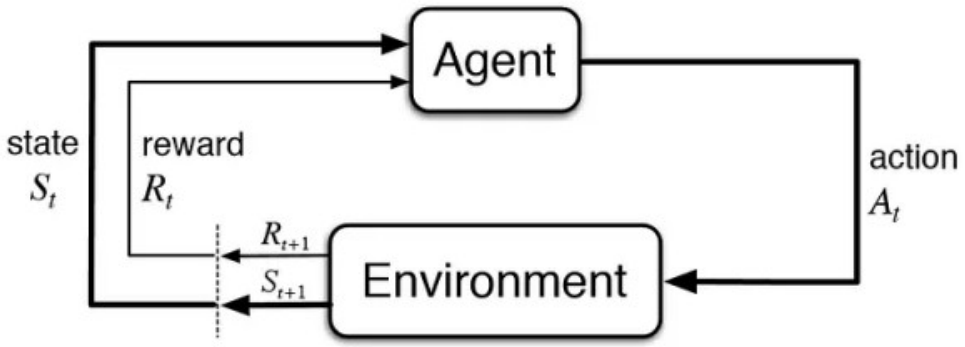
While Proportional-Integral-Derivative (PID) control is the industry standard for stable piloting. It offers a simple, white-box solution mentioned before. But it often struggles with complex situations where the Integral term alone is insufficient. Recently, Reinforcement Learning (RL) has gained traction as a robust black-box alternative. To reduce the computational burden on onboard systems, reinforcement learning has increasingly been adopted as a key approach for autonomous obstacle avoidance (Yu et al., 2025).

The concept of implementing RL in computing is among the earliest visions of AI.In a 1948 report, Alan Turing described a design for a pleasure-pain system (Barto, 2019). Modern RL defines a framework where an agent learns behavior through trial-and-error interactions with a dynamic environment (Kaelbling et al., 1996). Unlike supervised learning, the agent is not provided with correct answers. Instead, it receives a scalar reinforcement signal (reward or punishment) and must discover actions that maximize its long-term cumulative reward.

Markov Decision Processes (MDPs) provide the mathematical foundation for RL.In an MDP, an agent's actions influence both the immediate reward and the subsequent state of the environment. An MDP shown in Figure 5 is formally defined by (1) A set of states (S): Possible environment configurations; (2) A set of actions (A): Choices available to the agent;(3) A reward function (R): $R : S \times A \to \mathbb{R}$, specifying the instantaneous reward; (4) A state transition function (T): $T : S \times A \to \Delta(S)$, defining the probability of transitioning to state s' given action a in state s.

For training simulators, reinforcement learning can be used as a new way to build co-training agents.

Figure 5
Markov Decision Process diagram



Human–Computer Interaction in XR

Hardwares Introduction

Modern extended reality (XR) hardware can be classified based on the location of its compute unit into two categories: PC-tethered virtual reality (PCVR) systems and standalone (all-in-one) systems.

PCVR (Tethered): The headset acts as a peripheral. Assets are rendered on a high-end PC and streamed to the device via a wired (Link) or wireless (Air Link) connection. This allows developers to utilize the full power of engines like Unreal Engine, enabling high-fidelity features such as Lumen and Nanite. PC VR also supports specialized peripherals like haptic suits or omnidirectional treadmills, as seen in industry-standard titles like Half-Life: Alyx.

Standalone (All-in-One): Standalone headsets are self-contained systems with internal mobile chipsets. These provide high mobility but require significant optimization. High-end features must often be disabled or simplified to maintain performance. However, lightweight engines like Unity or Godot excel in this space, offering a balance between visual quality and mobile performance.

(Rendevski et al., 2022) note that Standalone VR (SVR) applications offer significant advantages over PC VR. With proper optimization techniques, almost any PC VR scenario can be developed and deployed as SVR while still providing an acceptable user experience. Differently,

a recent survey from (Game Developers Conference, 2026) shows that among developers interested in creating VR/AR/MR games, the SteamVR platform (PC VR) slightly surpassed the Meta Quest / Horizon Store (68% vs. 64%) as their platform of interest. These were followed by PlayStation VR/VR2 (34%) and Apple visionOS (15%). However, when filtering for developers who have actually worked on VR/AR/MR games, a strong preference remains for the Meta Quest (72%) over SteamVR (59%).

Due to hardware availability, this paper focuses exclusively on development for the Meta Quest 3.For Head-Mounted Devices (HMDs) like the Meta Quest, two primary architectural configurations exist:

Software Standards

To ensure cross-platform compatibility, developers rely on unified industry standards. Currently, there are two major public standards, OpenXR and WebXR.

OpenXR is the industry-standard API for XR applications. It is the interface between an application and an in-process or out-of-process XR runtime system. The runtime system handles functionality such as frame composition, peripheral management, and raw tracking information (Khronos OpenXR Working Group, 2024). Major game engines and the Meta Spatial SDK are built upon OpenXR to ensure hardware-agnostic development.

A primary bottleneck in XR development is the deployment loop. Compiling a standalone APK and pushing it to a device can take 10 15 minutes per iteration. There are two solutions to avoid the compiling: (1) Link Mode: Developers often use the Oculus Link App to preview scenes in PC VR, though this requires constantly putting on and taking off the headset. (2) Simulators: The Meta XR Simulator is increasingly becoming the preferred method for rapid iteration, allowing developers to simulate head and controller movements directly on their PC without needing physical hardware.

The WebXR Device API provides the interfaces necessary to enable developers to build compelling, comfortable, and safe immersive applications on the web across a wide variety of hardware form factors (World Wide Web Consortium, 2022).

WebXR serves as a higher-level abstraction that runs in the browser, combining older specifications from WebVR and WebAR. The WebXR standard shares core features with and can be viewed as a subset of OpenXR. While traditional game engines offer WebXR export support, there are fewer dedicated development stacks built exclusively for it.

For security reasons, WebXR APIs are restricted to HTTPS.Consequently, developers must use port-forwarding technologies during local development. Thanks to the efficiency of the web stack, a WebXR application typically only requires a few seconds to bundle. It is worth noting, however, that web rendering technology is currently transitioning from WebGL2 to WebGPU, which may introduce API changes in the WebXR ecosystem.

Although web technologies are widely used across various domains, WebXR remains an emerging technology and is still considered relatively unstable. Therefore, this study adopts the OpenXR standard to ensure greater stability and cross-platform compatibility.

XR Inputs

XR UI Frameworks

Digital Twin and Spatial Computing Concepts

Geographic Information Systems

Digital Twin

AI-assisted environment generation

Methodology

Design-Based Research

(Richardson et al., 2023) (Jong et al., 2017) (Wang et al., 2026)

Design-Based Research is an iterative, real-world research methodology that integrates design, implementation, and analysis to both improve educational interventions and generate theory.

Agile Software Development

(Turk et al., 2005) (Beck & Andres, 2004) (Schwaber & Beedle, 2001)

Simulate the Quadcopter

This section examines the flight dynamics and control implementation found in the Yue's Ultimate Drone physics model (Yuetility-Studios, 2022).

Controller Mapping

The flight controller distinguishes between unipolar throttle inputs and bipolar command sticks (Roll, Pitch, and Yaw): Throttle (T): Mapped to a normalized range of $[0,1]$. Attitude Commands (θ, ϕ, ψ): Mapped to a range of $[-1,1]$.

To refine the pilot's control over the aircraft, the system utilizes a non-linear control curve for joystick calculation. As (Lovesay, 2003) observes, a good quality force tracking controller has a linear output, with joystick force directly proportional to sight velocity, but many systems shape the joystick voltage output, to achieve fine control and greater tracking accuracy as the joystick output approaches zero, whilst retaining the same maximum output capability.

In this implementation, shaping is achieved by combining proportional (p) and exponential (e) components to calculate the target angular rate (τ) from the raw angular rate (u) as shown in Equation (7).

$$\tau = p \cdot u + e \cdot \mathrm{sgn}(u) \cdot u^2 \tag{7}$$

The raw gamepad throttle $u_T \in [-1,1]$ is normalized to a linear thrust scalar as shown in

Equation (8)

$$T = \frac{u_T + 1}{2} \tag{8}$$

Flight Modes and Orientation Logic

Since the Unity engine utilizes the FixedUpdate method for physics calculations, the drone's target orientation is updated discretely at every physics timestep (Δt).

In Acro mode, the joystick inputs command the angular rate of change. The drone's target orientation q_{target} is an accumulation of these rates over time. The orientation is updated by integrating the processed angular rates ($\tau_\phi, \tau_\theta, \tau_\psi$) as shown in Equation (9), the Quaternion.Euler equation is Equation (5).

$$\begin{gathered} q_\Delta = \text{Quaternion.Euler}(\tau_\theta \Delta t, \tau_\phi \Delta t, \tau_\psi \Delta t) \\ q_{\text{target},k} = q_{\text{target},k-1} \cdot q_\Delta \end{gathered} \tag{9}$$

In Self-Leveling mode, intended to prevent crashing and ease navigation, the pitch and roll sticks command a specific angle rather than a rate. Given a predefined maximum bank angle for pitch (θ_{max}) and roll (ϕ_{max}), the target values are calculated as Equation (10).

$$\theta_{\text{target}} = u_\theta \cdot \theta_{\text{max}}, \quad \phi_{\text{target}} = u_\phi \cdot \phi_{\text{max}} \tag{10}$$

The simulation applies two primary vectors to the drone's Rigidbody: appliedForce (Thrust) and appliedTorque (Attitude Control).

The total thrust force $\mathbf{F}_{\text{thrust}}$ is applied along the vehicle's local up vector ($\hat{\mathbf{u}}_{\text{up}}$). The magnitude is governed by the throttle T and the maximum thrust constant F_{max}, the thrust force is calculated as shown in Equation (11).

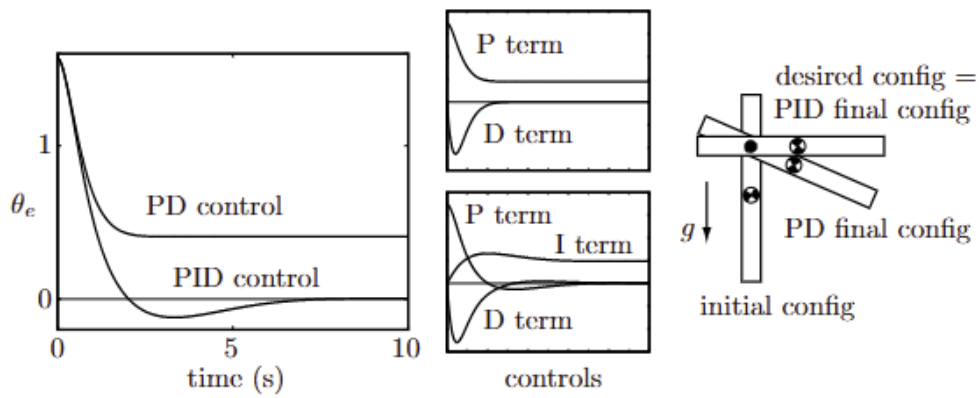
$$\mathbf{F}_{\text{thrust}} = (T \cdot F_{\text{max}}) \cdot \hat{\mathbf{u}}_{\text{up}} \tag{11}$$

Design of the Human-AI Collaboration

Design of a PD based HAC

Proportional integral derivative (PID) shown in Equation (6) can be applied for making a simple aviation helper in piloting the quadcopter. For basic flight stabilization, a simple PD action can provide satisfactory performance shown in Figure 6.

Figure 6
Comparison of PID and PD



For attitude control, The error between the target orientation q_{target} and the current body orientation q_{body} is defined as Equation (12).

$$\Delta q = q_{\text{target}} \cdot q_{\text{body}}^{-1} \tag{12}$$

Using Unity's ToAngleAxis method, this relative rotation is converted into an angle ϕ and an axis $\mathbf{n}$. The error vector $\mathbf{e}$ is then $\mathbf{e} = \phi \cdot \mathbf{n}$.

The controller applies a PD loop (Proportional-Derivative) to generate corrective torque shown in Equation (13).

$$\tau_{\text{world}} = K_p \cdot \mathbf{e} + K_d \cdot \frac{\mathbf{e} - \mathbf{e}_{k-1}}{\Delta t} \tag{13}$$

The world-space torque is transformed into the body frame (R_{wb}) for the physics

simulation: $\tau_{\text{body}} = R_{\text{wb}} \cdot \tau_{\text{world}}$

The controller should be able to pilot the quadcopter in a closed-loop mission. The system employs a cascaded control scheme where the Outer Loop (Position) provides setpoints for the Inner Loop (Attitude/Velocity).

The Inner Loop features controllers that yield the required rolling and pitching moments τ_x and τ_y to regulate horizontal movement. The Inner Loop block requires the current quadcopter attitude as well as reference signals for roll and pitch to obtain its output (Paredes et al., 2021); The Outer Loop, which features controllers that yield the required thrust differential and yawing moments as well as the desired reference signals.

During takeoff, the controller focuses exclusively on vertical error (e_y). The vertical force $forceY$ is mapped to a normalized throttle value (T) shown in Equation (8).

Following a path requires a look-ahead method. Given current position $\mathbf{p}$ and look-ahead target $\mathbf{t}$ on the path, the position error is defined as: $e_x = t_x - p_x, \quad e_y = t_y - p_y, \quad e_z = t_z - p_z$.

The Outer Loop converts this error into a desired velocity in the world axis as shown in Equation (14).

$$v_{x,\text{des}} = \text{PD}_X(e_x), \quad v_{z,\text{des}} = \text{PD}_Z(e_z) \tag{14}$$

To translate these into drone-relative commands (Pitch/Roll), the world velocity is projected onto the drone's Heading Axis: (1) Forward Basis: $\hat{\mathbf{f}} = R_y(\psi)\,\hat{\mathbf{z}}_{\text{forward}}$; (2) Right Basis: $\hat{\mathbf{r}} = R_y(\psi)\,\hat{\mathbf{x}}_{\text{right}}$; The projections shown as Equation (15). Where $e_{v,\|}$ is the longitudinal velocity error, $e_{v,\perp}$ is the lateral velocity error.

$$v_{\text{des}}^{\|} = \mathbf{v}_{H,\text{des}} \cdot \hat{\mathbf{f}}, \quad v_{\text{des}}^{\perp} = \mathbf{v}_{H,\text{des}} \cdot \hat{\mathbf{r}} \tag{15}$$

The Inner Loop then calculates the error between desired and actual velocity to produce stick inputs. Pitch (Right Vertical): Based on $e_{v,\|}$; Roll (Right Horizontal): Based on $e_{v,\perp}$.

The target yaw speed $\psi_{\text{target}} = \text{atan2}(v_x, v_z)$, the yaw error is $e_\psi = \text{DeltaAngle}(\psi, \psi_{\text{target}})$,

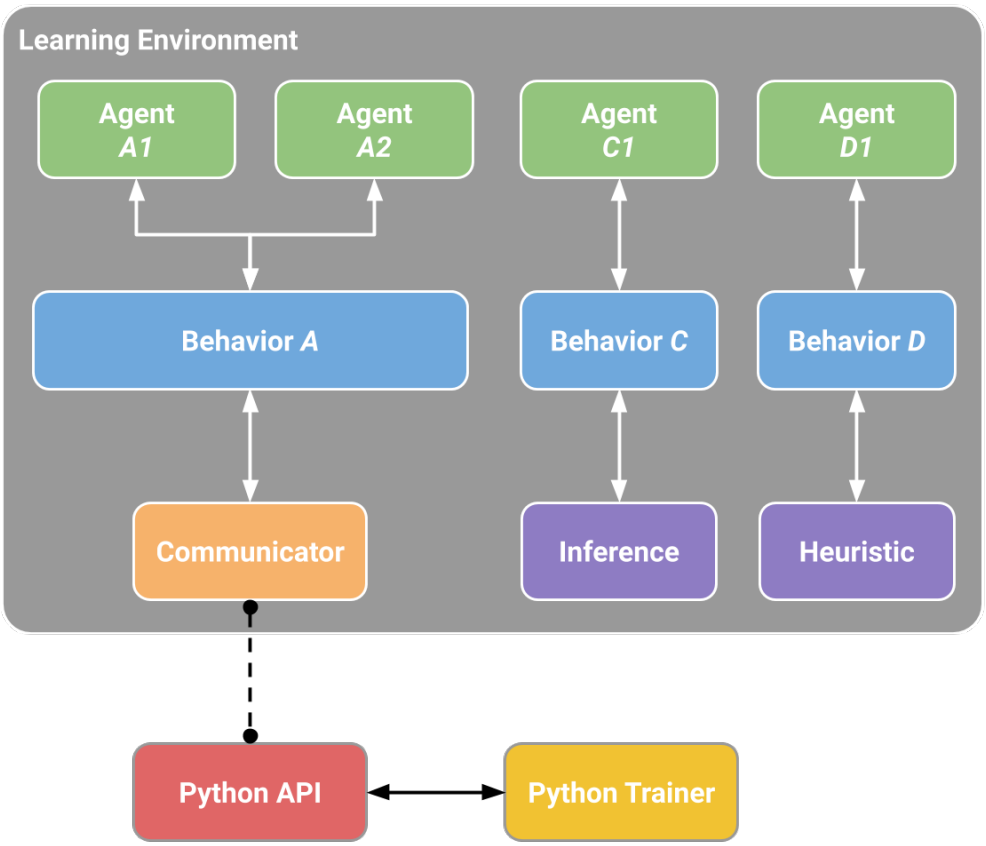
so mapped to joystick movement is $u_{\text{yaw}} = \text{PID}_{\text{yaw}}(e_\psi)$. The other axis are the same methods.

Design of a RL based HAC

Unity's ML-Agents Figure 7 enable games and simulations to serve as environments for training intelligent agents in Unity. Training can be done with reinforcement learning, imitation learning, neuroevolution, or any other methods. Trained agents can be used for many use cases, including controlling NPC behavior (in a variety of settings such as multi-agent and adversarial), automated testing of game builds and evaluating different game design decisions pre-release.

Every Learning Environment will always have one Agent for every character in the scene. While each Agent must be linked to a Behavior, it is possible for Agents that have similar observations and actions to have the same Behavior.

Figure 7
Simplified ML-Agents Scene Block Diagram



For this research, each Drone Agent includes, (1) Agent Script, Handles

OnActionReceived (processing joystick inputs), CollectObservations (monitoring states), and Heuristic (manual control for debugging); (2) Behavior Parameters: Defines the vector space for actions and observations. (3) Decision Requester: Controls the frequency of actions. By limiting the request rate, we can simulate more human-like reaction times, making the agent a better tutor for human pilots.

Proximal Policy Optimization (PPO) is an on-policy actor-critic algorithm that updates policies in small, clipped steps to ensure stability (OpenAI, 2017). PPO is particularly effective for drones learning to navigate complex environments (Yu et al., 2025).

The optimization objective is defined as Equation (16) shows.

$$L^{CLIP}(\theta) = \hat{\mathbb{E}}_t[\min(r_t(\theta)\hat{A}_t, \text{textclip}(r_t(\theta), 1-\varepsilon, 1+\varepsilon)\hat{A}_t)] \tag{16}$$

- θ is the policy parameter
- $\hat{E}_t$ denotes the empirical expectation over timesteps
- r_t is the ratio of the probability under the new and old policies, respectively
- $\hat{A}_t$ is the estimated advantage at time t
- ε is a hyperparameter, usually 0.1 or 0.2

Compared with a PD AI agent, a reinforcement learning-based AI agent does not require repeated testing to fine-tune parameters. Decisions are executed through a Decision Requester, which makes it easier to adjust the request frequency and pacing. However, as a black-box approach, it can be more difficult to achieve optimal performance or interpret and refine its behavior.

Comparison of Current XR Development Platforms

XR development is heavily dictated by the choice of game engine. Before initiating a project, it is critical to determine the target hardware and environmental requirements, as these factors define the development constraints.

Mixed reality (MR) headsets create immersive experiences designed to spatially integrate virtual content into the physical world. The newest headsets rely on passthrough video. While using passthrough, a person does not see light from the real world but instead relies on stereoscopic, color, high resolution, low latency, real-time video of the world, which is displayed on small screens inside a headset (Bailenson et al., 2024).

(Waldow et al., 2025) investigates Pass-Through Embodiment (PTE), The study found that PTE significantly enhances the user's sense of presence and embodiment compared to using a customizable digital avatar. The ability to see one's own body strengthens the feeling of being physically present in the VR environment without negatively affecting performance or causing sickness.

Leveraging passthrough on modern XR devices can help developers build highly engaging mixed-reality environments, which is particularly beneficial for applications like prototyping drone simulations.

Here, we present a comparison of current game engines that support extended reality, based on programming language, physics engine, rendering methods, Meta support, machine learning support, and editor size (see Table 2).

Table 2
Software Engine Comparison

Feature	Unreal Engine (5.7)	Unity (6.0)	Godot (4.6)	Wonderland Engine
Language	C++, Blueprints	C#	GDScript, C#	JavaScript, TypeScript
Physics	Chaos Physics	NVIDIA PhysX, DOTS	Jolt Physics	NVIDIA PhysX
Rendering	Lumen, Nanite	URP, HDRP	Forward+, Mobile	WebGL2
Meta Support	Fork build	First-class	Community-driven	No support
ML Support	Learning Agent	ML Agents	Godot RL Agents	No support
Editor Size	∼50GB+	∼20GB	$<$ 1GB	200MB

Unreal Engine: UE is known for photorealistic rendering, UE provides unparalleled visual

fidelity. Its Chaos Physics engine is highly regarded for complex simulations, making it the foundation for projects like Microsoft's AirSim. However, UE has a high barrier to entry: the editor is massive, and utilizing Meta's private APIs often requires compiling a custom engine branch from source.

Unity: Unity remains the most popular choice for Meta Quest development due to its mature SDK support and C# scripting. While it uses PhysX by default, its DOTS (Data-Oriented Technology Stack) provides high-performance physics for complex drone swarms. It offers the most streamlined out-of-the-box experience for Standalone VR.

Godot: Godot has emerged as a powerful, lightweight alternative. It is highly compatible with modern AI coding assistants due to its Python-like GDScript. The integration of Jolt Physics as the default 3D engine has resolved previous performance hurdles for VR. However, its XR ecosystem is largely community-maintained. While excellent for prototyping, the lack of official Meta API support and the potential instability of third-party GDExtensions require careful dependency management for production-grade projects.

Wonderland Engine: Wonderland Engine is a specialized WebXR engine built in C++ and compiled to WebAssembly. By leveraging WebGL2, it achieves the performance necessary to render thousands of objects at native frame rates directly within a browser. The development workflow relies on JavaScript and TypeScript, providing a highly familiar and accessible environment for traditional web developers. A unique advantage is the ability to edit scenes directly on the headset, enabling instant feedback without removing the device.

System Design Overview

Archetecture

Models

Environment Construction

Interaction and Control Design

Learning System Design

Data Collection Methods

Results

Control the Quadcopter in XR environment

The architecture of Yue's ultimate drone physics shown in Figure 8. A quad GameObject includes (1) Rigidbody, Unity's physics (2) YueInputModule, used to calculate controller axis. (3) YueDronePhysics, used to calculate the physics should applied on the drone's body with YueInputModule. (4) DroneXrEmulator, Map the Meta Quest Touch controller joystick data to YueInputModule.

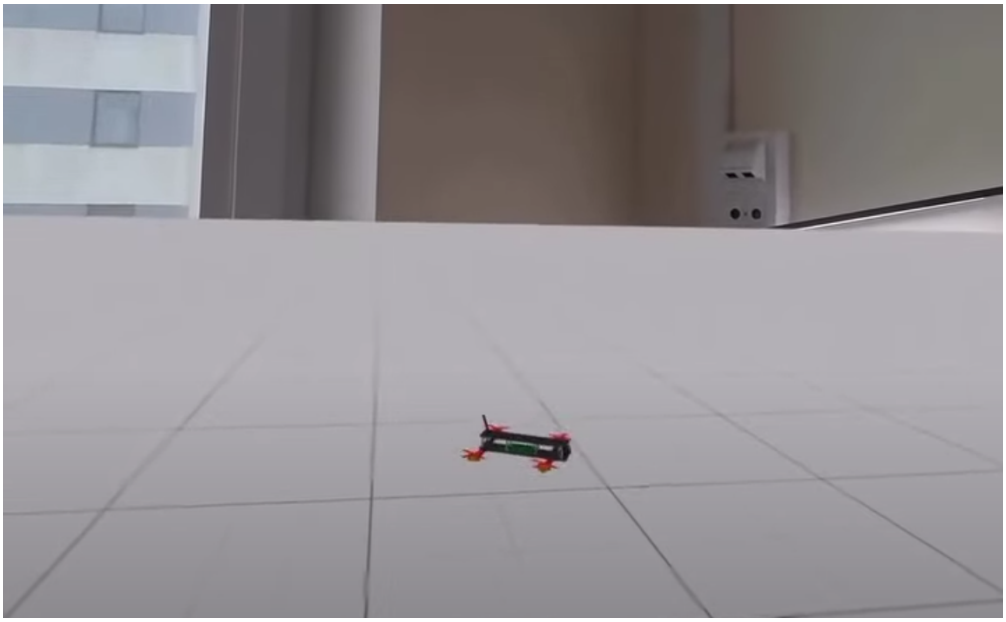
Figure 8
The Components of the Quad GameObject

GameObject : **Quad**
Rigidbody
YueInputModule
YueDronePhysics
DroneXrEmulator

To implement video passthrough, The scene should include the AR session into the scene, also add AR Raycast manager into the XR Origin. And add "AR Camera Manager" and "AR Camera background" on the MainCamera.

The result as shown in Figure 9

Figure 9
Controlling the drone physics in XR environment



Path Following with PID Controller

It is better to complete this section in a flat panel display. As this part is more algorithm centered. Instead of using the DroneXrEmulator in Figure 8, it is simpler to make another PidDroneEmulator which maps the calculated joystick value to YueInputModule.

To implement path following, two other components are included, PathFollower, cut the path into different targets for PidDroneEmulator to calculate errors and PathGenerator to build the path.

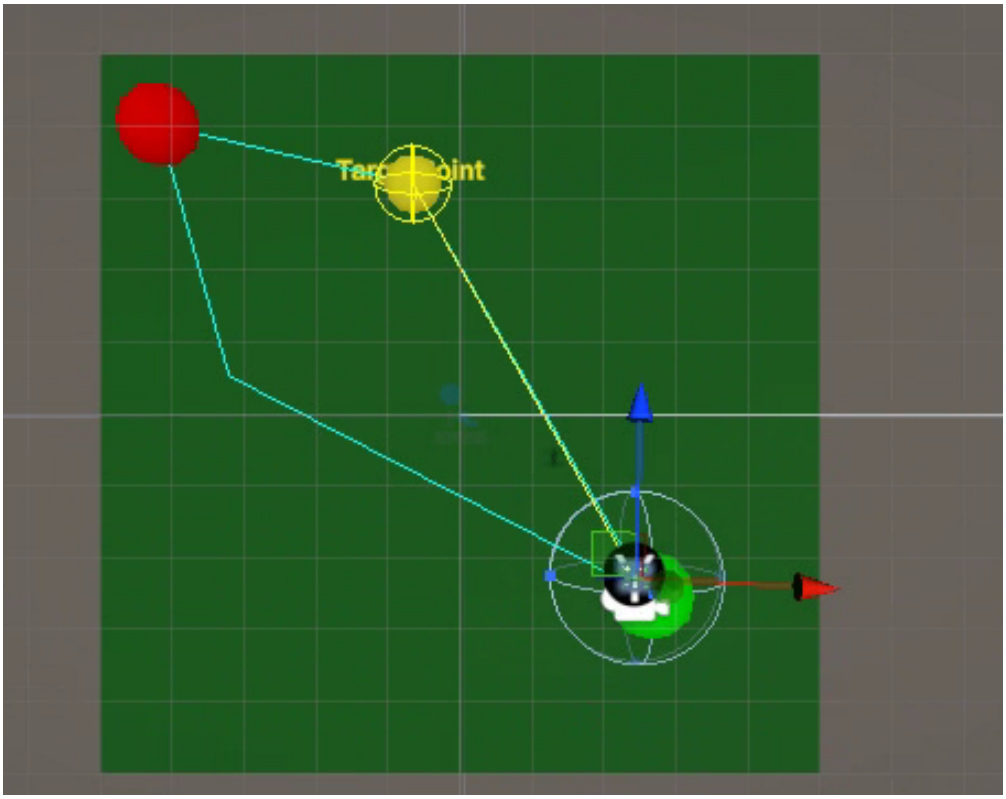
To move between points, the PathGenerator generates a path using two Cubic Bézier curves. Given four control points P_0, P_1, P_2, P_3, home to waypoint and back, the position $B(t)$ for $t \in [0,1]$ is defined as Equation (17).

$$B(t) = (1-t)^3 P_0 + 3(1-t)^2 t P_1 + 3(1-t) t^2 P_2 + t^3 P_3 \tag{17}$$

The path is constructed relative to the Home and Target vectors, the drone will fly from Home and fly back, the path as shown in Figure 10. Path from Home to Waypoint using a

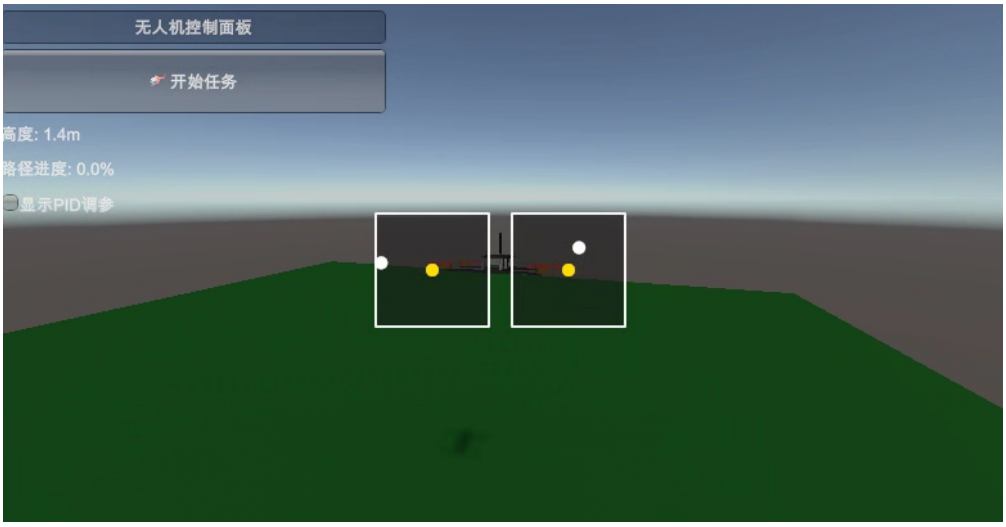
right-hand offset for control points; Path from Waypoint back to Home using a left-hand offset.

Figure 10
Path Design



By calculating these normalized joystick rates, the system can visualize the AI's calculation on the HUD.This serves as an AI collaboration hinter, showing human pilots the optimal stick positions for maintaining the designated path Figure 11.

Figure 11
hinter UI



Hovering with RL Agent

We implemented a 1D control mission: regulating the drone's vertical position (y) within a target band $[H-R, H+R]$, where H is the 'targetHeight' and R is the 'acceptableRadius'.

The following two values are being observed. (1) Normalized Height Error: $\text{clip}(\frac{y-H}{R}, -k, k)/k$ (where k is 'errorClampK'). This maps the error to a $[-1,1]$ range, preventing extreme spikes from destabilizing the gradient. (2) Vertical Velocity: v_y/v_{ref}, clipped to $[-1,1]$.

Rewards are calculated at each decision step (Δt): (1) Event-based: A firstArrivalReward is granted upon entering the target band for the first time per episode. (2) Dense/Steady-state: A livingRewardPerSecond is granted while maintaining position within the band. (3) Shaping: A penalty proportional to the normalized excess $(|e|-R)/R$ is applied when outside the band to discourage drifting.

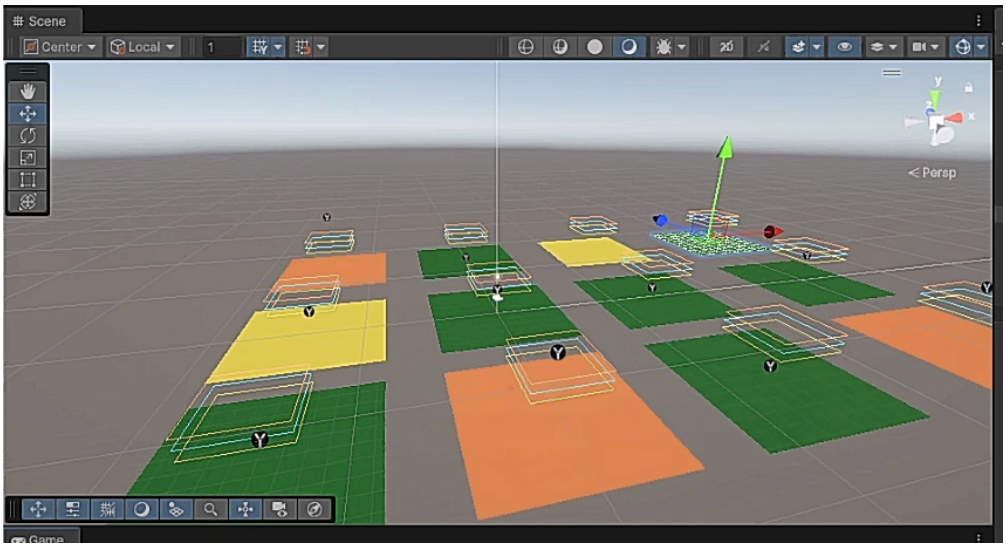
The agent was trained for 5×10^5 steps using the following parameters with Proximal Policy Optimization Equation (16) approach:

- Clipping parameter $\varepsilon = 0.2$
- GAE parameters $(\lambda, \gamma) = 0.99$
- Neural network: a 2-layer MLP with 128 hidden units

- Learning rate: 3×10^{-4} with linear decay

To monitor progress, we implemented a color-coded feedback system, Figure 12: Yellow: Below target; Orange: Above target; Blue: Stable hovering.

Figure 12
training process



From the train report, Cumulative Reward: Showed steady growth after 100k steps, plateauing at 400k, Figure 13. Episode Length: Positively correlated with rewards; as the agent improved, it survived longer without crashing (falling too low). Environment/Cumulative Reward Histograms: also shows when training time raises, reward start to raise.

The RL agent successfully learned to hover with high precision. By adjusting the DecisionRequester, we created a smoother control hint for the UI, matching the pace of human perception Figure 14.

Figure 13

Training result of the RL agent

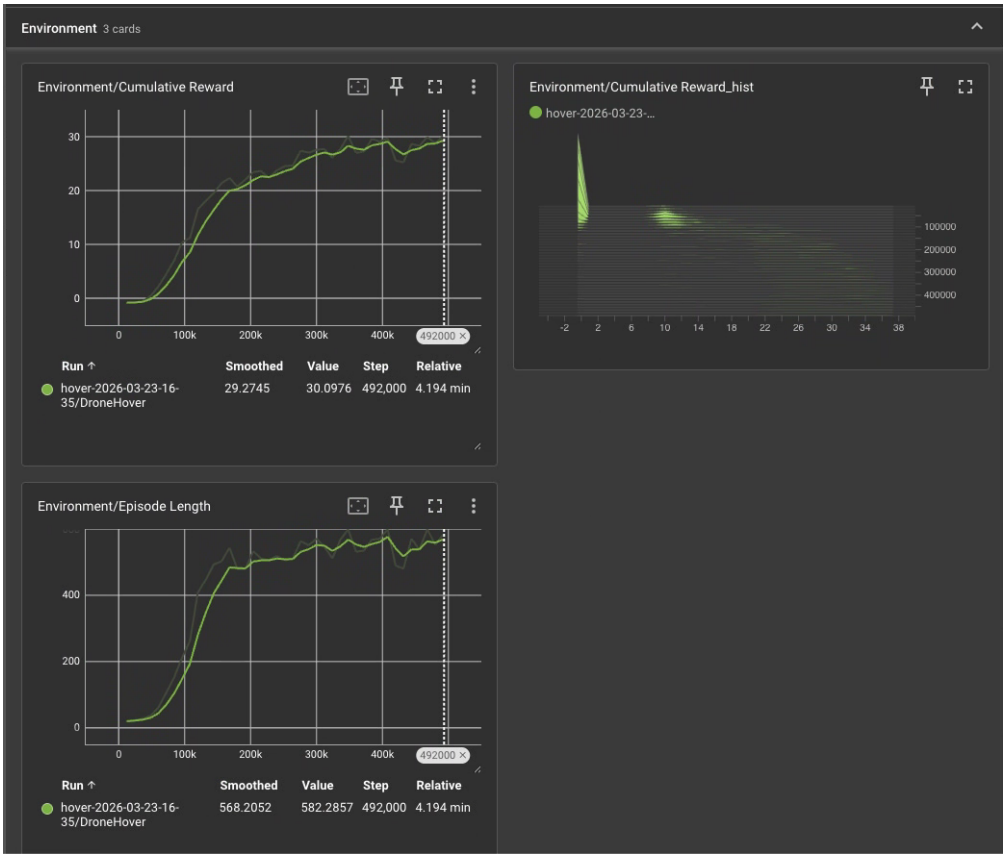
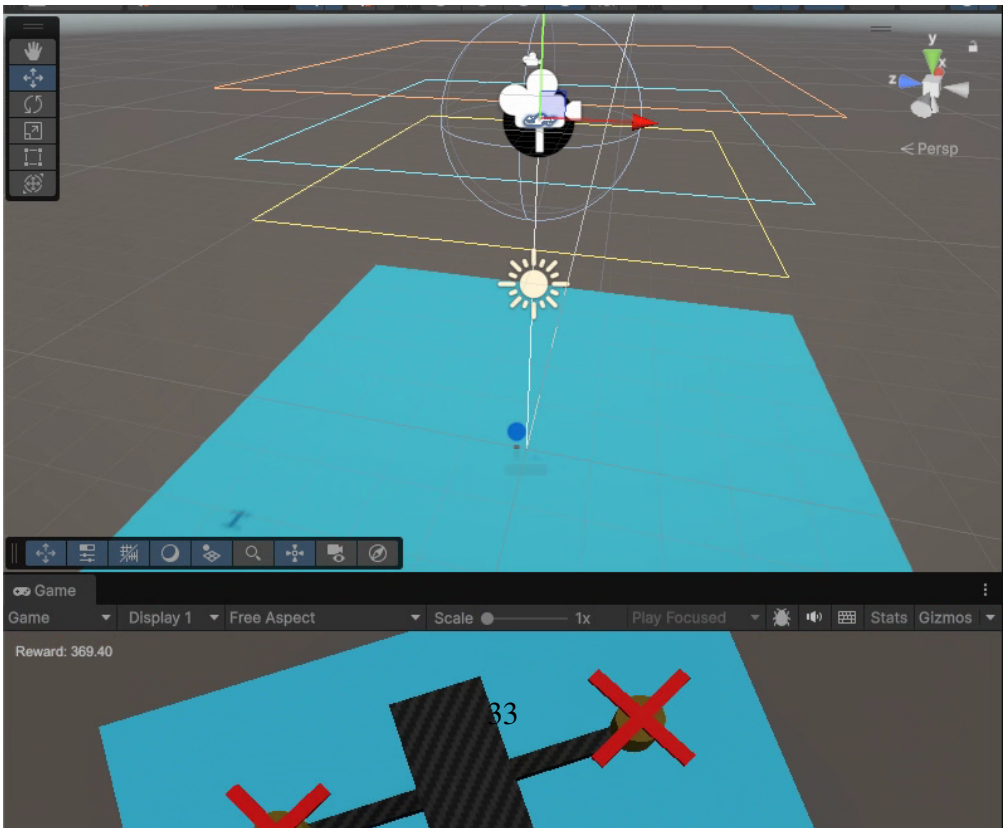


Figure 14

UI prototype of the RL learning agent



Tutorial on Game Interaction

Control the Quadcopter in VR environment

Drone Control Implementation

Physics and Simulation System

XR Interaction and UI Results

Discussion

RQ1: How Can FPV Quadcopter Flight Physics Be Simulated?

This study adopts Yue's Ultimate Drone physics model (Yuetility-Studios, 2022) as a reference implementation for simulating first-person view (FPV) quadcopter flight dynamics. The model employs a proportional-derivative (PD) control approach for flight stabilization, which is widely used in modern quadcopter systems.

In addition, a mapping strategy was developed to translate the controller profile of the Meta Quest into a simulated drone transmitter, thereby enabling intuitive user interaction within the virtual environment.

To ensure safe operation during simulation, the environment incorporates collider-based boundaries that constrain the quadcopter within a predefined area.

Comparison With Existing Simulators

FPV simulators (e.g., Liftoff, Uncrashed) primarily adopt cascaded proportional-integral-derivative (PID) control models to replicate real-world flight controllers. These systems typically support multiple controller configurations and provide interfaces for PID parameter tuning. However, such configurations can be difficult for beginners to set up with their own controllers.

Although most simulators provide a variety of selectable environments, they are primarily designed for flat-panel displays and do not offer the immersive learning enabled by extended reality (XR) environments.

Commercial FPV simulators commonly employ Acro (acrobat) mode to achieve realistic flight behavior. In contrast, this study focuses on a self-leveling flight mode, which constrains the drone's body angle and reduces the likelihood of crashes, thereby minimizing user frustration during early-stage learning.

Unlike existing simulators, the proposed system incorporates a human-AI collaboration mechanism that provides calculated control suggestions to support user decision-making.

Although Acro mode remains important for future investigation due to its higher realism,

it also introduces a greater likelihood of crashes. This raises several design considerations, including mechanisms for crash recovery, feedback systems for crash detection, and strategies for enabling AI agents to avoid collisions. From a game design perspective, it is also necessary to evaluate whether and how penalties should be applied following crashes.

RQ2: How can a human-AI collaboration agent be designed to assist users?

This study presents two approaches for designing a human-AI collaboration agent. The first approach is a transparent, white-box method based on proportional-integral-derivative (PID) control. The second approach is a black-box method using reinforcement learning (RL).

Comparison of PID and RL

To support user interaction and feedback, a heads-up display (HUD) was implemented to visualize the AI agent's real-time predictions of joystick inputs. The interface consists of two panels with crossed axes, where points represent the predicted joystick positions.

The results indicate that the RL-based solution provides greater flexibility during prototyping, as the frequency of agent decision-making can be adjusted dynamically. However, due to its black-box nature, effective fine-tuning remains challenging and represents an important direction for future research.

RQ3: What interaction methods enable high-fidelity XR prototyping?

This study employs the Mixed Reality Toolkit (MRTK) in combination with the Unity XR Toolkit to scaffold the user interface. Control inputs are implemented using the Meta Quest Touch Controller to emulate real-world quadcopter interaction. The user operates within a local-floor reference space, while a view-based heads-up display (HUD) provides real-time control status feedback.

To simulate first-person-view (FPV) goggles, Unity's RenderTexture is used to project the quadcopter's camera feed onto a virtual display panel. This design enables users to simultaneously perceive both the surrounding environment and the drone's onboard perspective, thereby supporting situational awareness and potentially reducing simulator sickness.

The system incorporates three interaction environments: (1) Additive Environment Blend,

an augmented reality setting in which users perform tutorial tasks within their physical surroundings; (2) a white-boxed playground, which serves as a controlled training environment for developing piloting skills; and (3) real city piloting, a virtual geographic environment that enables navigation within a simulated urban landscape. The design of these learning environments is grounded in experiential learning theory (Kolb, 1984), which conceptualizes learning as a cyclical process of experience, reflection, and experimentation.

When XR helps

This study positions XR as a spatial prototyping medium for FPV drone learning system design. XR enables direct manipulation of control mappings in a three-dimensional environment, reducing the need for abstract interface layout design and accelerating iterative testing of interaction strategies.

When XR is not preferable

When spatial arrangement is not part of the design problem, XR may introduce unnecessary complexity without improving simulation fidelity or development efficiency. In such cases, traditional simulation or parameter optimization approaches, such as PID tuning or reinforcement learning-based control refinement, are more efficient.

Limitations

Self-Leveling Flight Mode Only

For implementation simplicity, the current system is limited to self-leveling flight dynamics. Although Yue's Ultimate Drone physics model supports both self-leveling and acrobatic modes, this study focuses exclusively on the former to align with the design of the human-AI collaboration agents, which are optimized for stable flight control.

Consequently, acrobatic behaviors are not supported in the current implementation, which restricts the system's applicability to advanced flight scenarios.

Visual Feedback Only

Extended reality (XR) systems can support multimodal feedback. Additional modalities, such as haptic feedback and spatial audio, were considered in the design phase of the human-AI collaboration system, but they were not implemented in the current system.

This decision was driven by the additional design complexity associated with determining appropriate feedback types, intensity levels, and integration strategies. These aspects remain open research challenges for future work.

No Participant Testing

This study primarily focuses on system design and implementation rather than controlled user evaluation. As XR-based control interfaces are still relatively unfamiliar to many users, variability in prior experience may significantly influence task performance.

This makes it difficult to isolate learning effects from interface unfamiliarity, thereby limiting the interpretability of the user study results.

Future Work

Evaluation Methodology for XR-Based Drone Prototyping

Future work should develop more reliable evaluation methods for XR-based drone prototyping systems. In current user studies, participants may have very different levels of experience with XR devices, which can strongly affect their performance and learning speed.

To address this issue, future studies should consider designing evaluation protocols that reduce the influence of prior XR familiarity.

Extending to Acrobatic Flight Mode

Future work should extend the current self-leveling flight system to include acrobatic (acro) mode. While acro mode enables more expressive and realistic quadcopter control, it also introduces significantly higher control complexity and instability. This extension raises several research challenges, including crash detection, recovery mechanisms, and AI-assisted strategies.

From a game design perspective, it is also important to investigate circumstances, such as

penalties or reset mechanisms.

Designing Multimodal Feedback Mechanisms

Future research should explore the integration of multimodal feedback, including audio, and haptic channels, to enhance user awareness during flight control. Rather than simply adding additional feedback modalities, it is important to investigate their placement, timing, and intensity, as these parameters may significantly affect cognitive load and learning performance. A systematic study of feedback design space could provide guidelines for more effective XR interaction design.

Reinforcement Learning for Path-Following Control

While hovering control can be effectively modeled using distance-based feedback signals, path-following tasks require reasoning over multiple spatial and temporal variables. Manually designing these features can become inefficient and non-scalable.

Future work should explore reinforcement learning approaches to automatically learn control policies from richer state representations. This may enable a transition from target-based navigation to more general vision-based flight control systems.

Imitation Learning for Human-Guided Control

Although imitation learning provides a promising direction for human-aligned control policies, it requires large-scale and high-quality human demonstration data. In the context of XR drone interaction, collecting such data introduces challenges in consistency, labeling, and user variability.

Future research should investigate efficient data collection strategies, including semi-automated annotation or human-in-the-loop learning, to reduce the overhead of dataset construction and improve training efficiency.

Summary

Goals

Develop XR training system with AI guidance.

Results

Functional system with immersive interaction.

Generalization

XR supports skill training.

Practical Use

Prototype training framework.

Future Study

Learning validation and system expansion.

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Appendix